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Where I Becomes Infinite

Few-Shot Prompting

Module 2.3: Prompt Engineering | Prof. Sashikumaar Ganesan | IISc Bengaluru

Session Overview

What You Will Learn

- What few-shot prompting is and how it differs from one-shot
- The research foundations of in-context learning
- How to select and order few-shot examples
- Trade-offs: accuracy vs. token cost

Session Details

- **Duration:** 8 minutes
- **Bloom's Level:** B3 — Apply
- **Difficulty:** L2 — Intermediate
- **Module:** 2.3 Prompt Engineering
- **Prereq:** One-Shot Prompting

Learning Objectives

By the End of This Session You Will Be Able To

- **Define** few-shot prompting and explain in-context learning (ICL)
- **Apply** few-shot prompting to classification and generation tasks
- **Select** representative, diverse, and correctly-ordered examples
- **Analyze** the accuracy-cost tradeoff when varying the number of shots

Few-Shot Prompting: Definition

Core Definition

Few-shot prompting provides the model with *two to ten* input-output examples before presenting the target task. Each example demonstrates the desired response pattern, enabling the model to generalize to new inputs with higher reliability and consistency than zero-shot or one-shot approaches.

Foundational Research

Introduced in the GPT-3 paper (Brown et al., 2020) as a key capability of large language models. Few-shot performance at scale rivaled fine-tuned models on many NLP benchmarks without any gradient updates.

The Prompting Spectrum

Zero to Many Shots

- **Zero-shot:** No examples. “Classify the sentiment of: [text]”
- **One-shot:** One example. Shows format, not pattern breadth.
- **Few-shot:** 2–10 examples. Demonstrates pattern across multiple cases.
- **Many-shot:** 10+ examples. Approaches fine-tuning effectiveness; bounded by context window size.

More shots improve consistency and coverage of edge cases, but consume more context window tokens and increase latency and cost.

Anatomy of a Few-Shot Prompt

Complete Structure

- **Instruction:** "Classify the emotion in each message: Joy, Anger, or Sadness."
- **Example 1:** Input: "I got promoted today!" $\rightarrow$ Emotion: Joy
- **Example 2:** Input: "The project was cancelled again." $\rightarrow$ Emotion: Sadness
- **Example 3:** Input: "They changed my code without asking!" $\rightarrow$ Emotion: Anger
- **New Input:** Input: "My team celebrated my birthday at work." $\rightarrow$ Emotion:

Why Multiple Examples Improve Performance

Statistical Coverage

- Multiple examples reduce the chance the model misinterprets the task
- They cover a broader range of input variability
- They reinforce the desired output format across different inputs
- They calibrate the model’s decision boundary for classification

In-Context Learning Mechanism

- Model attends to all examples simultaneously
- Pattern extraction is implicit, not explicit rule-following
- More examples = more reliable pattern signal
- Even incorrect labels can improve structure learning

How to Select Few-Shot Examples

Selection Principles

- **Diversity:** Cover different sub-cases of the input space (positive/negative/edge)
- **Representativeness:** Each example should reflect a real, common scenario
- **Correctness:** All labels must be accurate; wrong examples degrade performance
- **Balanced:** For classification, include roughly equal examples per class
- **Similar length:** Example outputs should match the expected output length

Does Example Order Matter?

Research Findings

- Yes. Research shows LLMs exhibit **recency bias**: the last few examples in the prompt have disproportionate influence on the output.
- Placing the most representative example last tends to improve performance.
- Randomly shuffling examples reduces order-induced bias across many queries.

Practical Recommendation

For production systems, randomly sample and shuffle few-shot examples at runtime to reduce order sensitivity and improve robustness across varied inputs.

Accuracy vs. Token Cost Trade-Off

Shots	Accuracy	Token Cost	Best Use Case
0 (Zero)	Moderate	Minimal	Common tasks, clear instructions
1 (One)	Good	Low	Format-sensitive tasks
3–5	High	Moderate	Complex classification
8–10	Very high	High	Near-fine-tune quality needed

Diminishing returns beyond 5–8 examples for most tasks. Context window limits apply.

Application: Medical Coding and Classification

Healthcare Revenue Cycle Automation

Healthcare providers use few-shot prompts to assign ICD-10 medical codes to clinical notes. A few examples covering common diagnosis types allow the model to generalize to new notes without training a specialized model.

Why Few-Shot Suits This Domain

- Medical coding vocabulary is specialized; a few examples establish domain context
- Output must exactly match a code from a controlled vocabulary
- Few-shot achieves near-human accuracy for common code categories
- Companies: Nuance, Cohere Health, AWS HealthLake

Application: Customer Intent Detection

Conversational AI Routing

Customer service platforms (Salesforce Einstein, ServiceNow AI, Genesys) classify customer queries by intent (billing, technical support, cancellation) to route them correctly. Few-shot prompts enable rapid deployment of new intent categories without retraining classifiers.

Few-Shot Intent Classification

- 3–5 examples per intent class establish decision boundaries
- New intents can be added by writing examples, not retraining
- Handles ambiguous queries better than rule-based routers

Application: Legal Clause Extraction

Contract Analysis at Scale

Legal tech firms (Ironclad, ContractPodAi, Harvey AI) use few-shot prompts to extract specific clause types (limitation of liability, indemnification, termination rights) from contracts. A few annotated examples enable extraction of new clause types in hours rather than months of model training.

Few-Shot Legal Extraction

- Examples cover diverse clause phrasings across different contract styles
- Output format (JSON with clause text + type) demonstrated in examples
- Performance scales with example quality, not just quantity

Application: Domain-Specific Code Generation

Custom Coding Standards Enforcement

Software teams use few-shot prompts to enforce company-specific coding patterns. By providing examples in the house style (variable naming, error handling, logging), the model generates code matching internal conventions without fine-tuning.

Few-Shot Code Patterns

- 3–4 examples show the team’s error handling conventions
- Model generates new functions following the same pattern
- Faster than writing detailed style guides in instructions alone

Key Takeaways

Core Concepts to Remember

- Few-shot prompting provides 2–10 examples to guide model behaviour via in-context learning
- More examples improve accuracy but consume context window tokens and increase cost
- Example selection: diverse, representative, balanced, and all correct
- Order matters: recency bias exists; shuffle examples in production for robustness
- Few-shot enables rapid deployment of new tasks without model retraining

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